1. The products Table

1. Classify each product by price into the three tiers below.

```
1.SELECT
2. Product_name, price,
3 CASE
WHEN price>1000 THEN 'Expensive'
WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
ELSE 'Budget'
END AS price_category
FROM products;
```

Output

Product_name	Price	Price_ category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	360.00	Mid-range
Mouse	25.00	Budget

2. Label each order by its value.

```
SELECT
customer_name, amount,
CASE
WHEN amount >= 1000 THEN 'High Value'
WHEN amount >= 500 THEN 'Medium Value'
ELSE 'Low Value'
END AS order_value_category
FROM orders;
```

Output

customer_name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	Low value
Ethan	1200.00	High value

3. Categorize each employee's position level using both department and salary

```
SELECT
emp_name, department, Salary,
CASE
WHEN department = 'IT' AND salary > 80000 THEN 'Senior
IT'
WHEN department = 'HR' AND Salary > 55000 THEN '
Experienced HR'
ELSE 'staff'
END AS position_level
FROM employees;
```

Output

emp_name	department	Salary	Position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. Assign each student a letter grade based on their score.

```
SELECT
student_name, score,
CASE
WHEN score >= 90 THEN 'A'
WHEN score >= 80 THEN 'B'
WHEN score >= 70 THEN 'C'
WHEN score >= 60 THEN 'D'
ELSE 'F'
END AS grade
FROM students;
```

Output

Student	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. label delivery performance based on the time taken.

```
SELECT
delivery_id, delivery_time_minutes,
CASE
WHEN delivery_time_minutes <= 30 THEN 'Fast'
WHEN delivery_time_minutes <= 60 THEN 'On Time'
ELSE 'Late'
END AS performance
FROM deliveries
```

Output

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6 Convert the numeric priority into a readable label.

```
SELECT
issue_type, priority,
CASE
WHEN priority = 3 THEN 'High'
WHEN priority =2 THEN 'Medium'
WHEN priority = 1 THEN 'Low'
END AS priority_label
FROM tickets;
```

Output

issue_type	priority	priority_label
login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7 Calculate attendance percentage and Classify the result.

```
SELECT
student_id,
(days_present * 100.0 / total_days) AS attendance_
percentage,
CASE
```

```
WHEN (days_present * 100.0 / total_days) >= 90 THEN 'Excellent'
WHEN (days_present * 100.0 / total_days) >= 75 THEN 'Good'
ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance
```

Output

Student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

8. Label the stock status of each product.

```
SELECT
product_id, stock_qty,
CASE
WHEN stock_qty = 0 THEN 'Out of stock'
WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low stock'
ELSE 'In stock'
END AS stock_status
FROM products_inventory;
```

Output

product_id	Stock_qty	Stock_status
1	5	Low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

~~10. Apply a discount flag based on the payment method and amount.~~

9. Classify each class by the number of enrolled students.

```
SELECT
Subject, enrolled_students,
CASE
WHEN enrolled_students > 25 THEN 'Large'
WHEN enrolled_students BETWEEN 10 AND 24 THEN
'Medium'
ELSE 'Small'
END AS Class_size_category
FROM Classes.
```

Output

Subject	enrolled_students	Class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10. Apply a discount flag based on the payment method and amount

```
SELECT
payment_id, payment_method, amount,
CASE
WHEN payment_method = 'Cash' AND amount >=
200
THEN 'Eligible for Discount'
ELSE 'Not Eligible'
END AS discount_eligibility
FROM payments;
```

Output

payment_id	payment_method	amount	amount	discount_eligibility
1	Card	50.00		Not Eligible
2	Cash	200.00		Eligible for discount
3	Card	150.00		Not Eligible
4	PayPal	75.00		Not Eligible
5	Cash	300.00		Eligible for discount